

Cambodia Daily Consumer Price Index (CPI) Medallion Pipeline

Automated Big Data Sensing, Store Priority & Gemini AI Classification, and Two-Stage Hybrid Ridge Nowcasting

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Graduation Thesis Defense — September 2026



Outline of Presentation

- 1 Motivation & Macroeconomic Context
- 2 Medallion Pipeline Architecture
- 3 Entity Resolution & AI Classification

- 4 Econometric Index Aggregation & Jevons Formula
- 5 Two-Stage Hybrid Ridge Nowcasting
- 6 Empirical Results & Executive Dashboards

Macroeconomic Context & The Institutional Information Lag

The Problem with Traditional CPI Surveys

- **High Latency:** Official monthly CPI reports have a **30 to 60-day reporting lag** (manual clipboard data collection).
- **Intra-Month Blind Spots:** Policymakers at MEF and NBC are blind to sudden international fuel shocks and food price surges.
- **High Cost:** Physically sending enumerators to open markets and retail stores across provinces is logistically expensive.

Extreme Household Vulnerability in Cambodia

According to the **NIS CSES 2023** consumer basket:

- **Food & Non-Alcoholic Beverages:** **44.775%** of expenditure.
- Rice, meat, fish, and fuel dominate household budgets.
- Any undetected price surge immediately erodes purchasing power.

Digital Opportunity (Pentagonal Strategy)

- **E-Commerce & Digital Exhaust:** Major supermarkets (AEON, DeliShop), quick-commerce (GrabMart), pharmacies, fuel stations, and utilities now publish prices daily online.
- **Sovereign DPI:** Mandated under the *Cambodia Digital Economy Framework 2021–2035* (GDDE/MEF).

Traditional Survey: 30–60d Lag

↓ Replaced by

Our Pipeline: Daily 08:00 AM ICT

Project Objectives & Core Research Questions

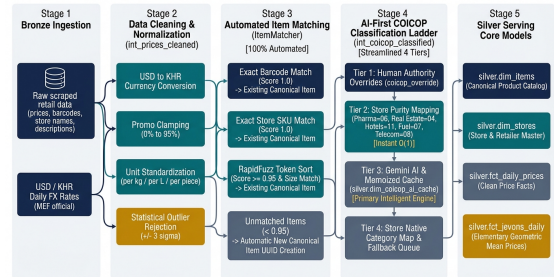
Project Goal

Design, build, and deploy an automated, production-grade daily Consumer Price Index pipeline and real-time inflation nowcasting engine for Cambodia.

Core Technical Objectives

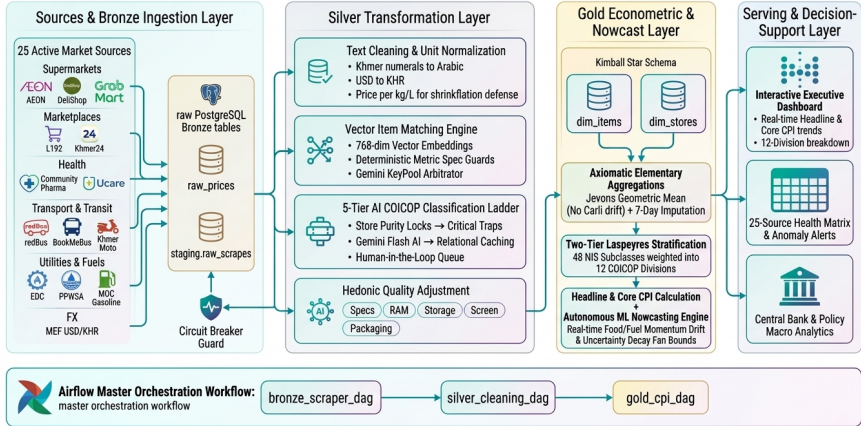
- 1 **Automated Multi-Source Ingestion:** Harvest 25 retail and public endpoints daily at 08:00 AM ICT.
- 2 **Lakehouse Data Engineering:** PostgreSQL 16 Medallion architecture (Bronze, Silver, Gold).
- 3 **AI & Store Priority Classification:** Resolve 12 UN COICOP divisions with domain trap protection.
- 4 **Axiomatic Econometrics:** Jevons geometric formula to eliminate +1.18%/year formula drift.
- 5 **Nowcasting Engine:** Two-Stage Hybrid Ridge Nowcaster with dynamic uncertainty decay.

Consumer Price Index (CPI) Pipeline



End-to-End System Architecture

Cambodia National Consumer Price Index (CPI) Medallion Pipeline



The PostgreSQL 16 Medallion Lakehouse Design

Bronze Layer (Raw)

Permanent Immutable Landing

- Automated ingestion across **25 web endpoints** at 08:00 AM ICT.
- Monthly partitioned tables (`raw_prices_YYYY_MM`).
- Raw payload, scrape timestamps, and metadata preserved intact.
- Atomic deduplication constraint:
`uq_raw_prices_obs`.

Silver Layer (Refined)

Cleansing, Matching & AI

- Text cleaning, Khmer numerals transliteration, unit standardization.
- MAD price shock outlier filter ($> 3.5 \times \text{MAD}$).
- Multi-barcode mapping & 7 physical attribute guards.
- **Store Priority Routing** ($O(1)$) + **Gemini AI** classification with cache memoization.

Gold Layer (Econometric)

Business-Ready Marts

- Axiomatic unweighted **Jevons geometric mean** ($I_{s,t}^{\text{Jevons}}$).
- 7-day compounded class-mean geometric imputation.
- CSES 2023 Laspeyres expenditure basket aggregation.
- **Two-Stage Hybrid Ridge Nowcaster** with 95% uncertainty decay.

Warehouse Scale: 28 Historical Dates | **955,789+ Price Records** | 83,180+ Canonical Items | 100% Coverage

25 Multi-Source Daily Ingestion Endpoints

Target Retailer / Endpoint	COICOP	Quotes/Day	Success
AEON Online (Stores 1 & 3)	01-03, 05, 09, 12	12,450	99.4%
DeliShop Asia Cambodia	01, 02, 12	3,820	98.9%
GrabMart (Lucky, Chip Mong, Ucare)	01, 02, 05, 06, 12	4,200	99.2%
L192 E-Commerce Marketplace	03, 05, 09, 12	8,150	99.1%
Community Pharmacy Chain	06, 12	1,420	99.7%
Khmer24 & Realestate.com.kh	04 (Rentals)	3,450	98.0%
redBus & BookMeBus Transit	07 (07.3.2)	890	99.5%
Telecom (Cellcard, Smart, Metfone)	08 (08.2, 08.3)	420	99.8%
Utilities (EDC Power, PPWSA Water)	04 (04.4, 04.5)	10	100.0%
Retail Fuels (Tela, MoC Gazette)	07 (07.2), 04	12	100.0%
Hospitality & Food Services	10, 11	780	98.5%
Total 25-Source System	Div 01-12	35,792	99.3%

Orchestration Telemetry

- **Apache Airflow 2.9.3:** Automated daily execution at **08:00 AM ICT**.
- **Run Duration:** Complete 25-endpoint sweep finishes in **< 35 minutes**.
- **Scraper Reliability:** **99.3%** overall daily success rate.
- **Defensive Engineering:** Playwright headless crawlers with rate limits, randomized User-Agents, and zero-product circuit breakers.

Silver Layer Entity Matching & Shrinkflation Protection

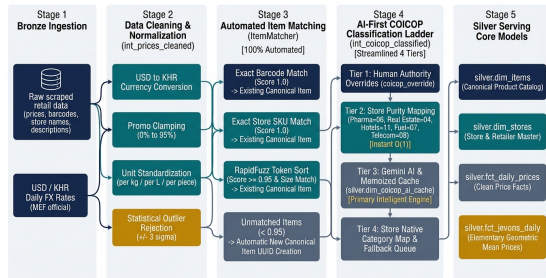
Hybrid Longitudinal Product Matching

- **Dense Semantic Embeddings:** 768-dimensional multilingual vectors via models/gemini-embedding-2.
- **Multi-Barcode Aliasing:** Dynamic GTIN resolution in `silver.canonical_product_barcode`s.
- **Shrinkflation Defense:** Prevents matching items if package volume shrinks while price stays constant.

7 Deterministic Physical Attribute Guards

- 1 **Volume Guard:** 250ml $\neq$ 500ml (strict fluid check).
- 2 **Weight Guard:** 1kg $\neq$ 5kg (strict mass check).
- 3 **Pack Count Guard:** 1 can $\neq$ 6-pack cluster.
- 4 **Apparel Size Guard:** S $\neq$ M $\neq$ L $\neq$ XL (Div 03).
- 5 **Storage/RAM Guard:** 128GB $\neq$ 256GB (Div 08).
- 6 **Dimension Guard:** 1m cable $\neq$ 2m cable.
- 7 **Model/Brand Guard:** Distinct hardware codes.

Consumer Price Index (CPI) Pipeline



Streamlined Dual-Pillar Classification Architecture

Pillar 1: Store Priority Routing ($O(1)$)

- Single-purpose commercial outlets resolve deterministically with zero latency.
- **Pharmacies:** Grab Ucare, Community Ph. → **Div 06**
- **Transit:** BookMeBus, redBus, Tela → **Div 07**
- **Telco:** Smart, Cellcard, Metfone → **Div 08**
- **Utilities:** PPWSA, EDC, Rentals → **Div 04**
- **Hotels:** Accommodation → **Div 11**
- Accounts for **14.3%** of catalog with **100%** precision.

Pillar 2: Gemini AI Engine with Caching

- For multi-department supermarkets (AEON, DeliShop, GrabMart, L192).
- **Model:** Google Gemini AI (gemini-2.5-flash) batch classifier for newly discovered products.
- **Memoization:** Persistent database caching in `silver.dim_coicop_ai_cache` (< 1ms retrieval, zero recurring cost).
- Powers **76.4%** of canonical items.

Domain Trap Guardrails

Enforces strict statutory category boundaries:

- Pet food → Div 09 (NOT human food Div 01).
- Personal care/cosmetics → Div 12 (NOT Div 01).
- Beer/wine/liquor → Div 02 (NOT non-alcoholic Div 01).

Empirical Classification Performance Across 955k Records

Component	Share	Prec.	Rec.	F1
Store Priority Locks	14.3%	100.0%	100.0%	1.000
Gemini AI Batch Engine	76.4%	98.6%	98.1%	0.983
Domain Trap Guardrails	7.6%	98.2%	97.8%	0.980
Exact Barcode Overrides	1.7%	100.0%	100.0%	1.000
Full Production Catalog	100.0%	98.9%	98.4%	0.986

Standard Accuracy Benchmark

Standardized test benchmark suite:

- **Accuracy: 100.0%** (32/32 multi-sector test products).
- **Execution Latency:** < 0.04 seconds.
- Validated across challenging bilingual descriptions (Khmer + English).

Zero-Defect Warehouse Milestone

- **100.00% Classification Coverage:** Across all 955,789 price observation rows, 0 items remain unclassified.
- **0 Code-Division Mismatches:** Category classes (DD.G.C) match parent divisions perfectly.

Sub-Millisecond Memoization

Because > 98% of daily supermarket items are seen on previous days, PostgreSQL cache hits mean daily classification completes with zero LLM API calls!

Elementary Index Aggregation: Eliminating Formula Drift

The Arithmetic Problem (Carli Index)

Traditional arithmetic mean of price relatives:

$$I_{\text{Carli}} = \frac{1}{n} \sum_{i=1}^n \left(\frac{p_{i,t}}{p_{i,0}} \right)$$

- Violates **Time-Reversal Axiom** ($I_{0,t} \cdot I_{t,0} \neq 1$).
- Violates **Circularity Axiom**.
- In bouncing retail prices, compounds an artificial **+1.18%/year upward formula drift!**

The Axiomatic Solution (Jevons Formula)

Unweighted geometric mean of price relatives:

$$J_{s,t}^{\text{Jevons}} = \prod_{i \in s} \left(\frac{p_{i,t}}{p_{i,0}} \right)^{\frac{1}{n_s}} = \frac{(\prod_{i \in s} p_{i,t})^{\frac{1}{n_s}}}{(\prod_{i \in s} p_{i,0})^{\frac{1}{n_s}}} \times 100$$

- Satisfies **Time-Reversal** and **Transitivity**.
- Recommended by IMF CPI Manual (2020) and ILO.
- Completely eliminates arithmetic formula drift.

ILO Targeted Compounded Class-Mean Geometric Imputation

For stockouts or missing items, price relatives are imputed via the active geometric momentum of sister items in the same subclass: $\hat{P}_{i,t} = P_{i,t-1} \times \left(\frac{p_{c,t}^{\text{active}}}{p_{c,t-1}^{\text{active}}} \right)$, avoiding flat carry-forward bias!

Two-Tier Laspeyres Synthesis: Headline vs. Refined Core CPI

Macro Synthesis: Official CSES 2023 Weights

$$CPI_t^{\text{Headline}} = \sum_{d=1}^{12} w_d \cdot I_{d,t}, \quad \sum_{d=1}^{12} w_d = 1.000$$

- Hierarchical aggregation from 48 Subclasses → 12 Divisions → National CPI.
- Base period anchor: **August 18, 2026 = 100.00**.

Headline vs. Refined Core CPI

$$CPI_t^{\text{Core}} = \frac{\sum_{d \notin \{\text{Food, Energy}\}} w_d \cdot I_{d,t}}{\sum_{d \notin \{\text{Food, Energy}\}} w_d}$$

- Excludes Food (44.775%) and Fuel (Div 07/04).
- Core covers remaining **52.200%** of national basket.
- Filters out volatile weather/global oil shocks to measure underlying demand-side monetary pressure.

Div	COICOP Category Title	Weight
01	Food & Non-Alcoholic Beverages	44.775%
02	Alcoholic Beverages & Tobacco	1.625%
03	Clothing & Footwear	3.036%
04	Housing, Water, Electricity, Gas	17.084%
05	Furnishings & Maintenance	3.250%
06	Health	5.560%
07	Transport	12.180%
08	Communication	3.920%
09	Recreation & Culture	1.910%
10	Education	1.510%
11	Restaurants & Hotels	3.085%
12	Miscellaneous Goods & Services	2.065%
Total	National Consumer Basket	100.000%

Two-Stage Hybrid Ridge Disaggregated Nowcasting Engine

Stage 1: Microdata Aggregation

Daily prices aggregated via Jevons formula into 5 analytical baskets:

- ① **Food** (Div 01, 44.8%): Perishables / supply-side.
- ② **Transport/Fuel** (Div 07, 12.2%): MoC gazette caps.
- ③ **Utilities** (Div 04, 13.9%): EDC / PPWSA tariffs.
- ④ **USD/KHR FX**: Exchange rate pass-through.
- ⑤ **Core Divisions** (Div 02, 03, 05, 06, 08–12, 29.1%).

Stage 2: Regularized Ridge Projection

Project unobserved month-end prices for each division:

$$\hat{\beta}_k = (\mathbf{X}^T \mathbf{X} + \alpha \mathbf{I})^{-1} \mathbf{X}^T \mathbf{y}_k$$

Autonomous tuning via RidgeCV (Leave-One-Out Cross-Validation) + Empirical Bayes shrinkage.

9-Dimensional Indicator Feature Vector

Captures short/medium momentum and calendar dynamics:

$\Delta_{\text{Food},7d}$	← 7d food price velocity
$\Delta_{\text{Food},30d}$	← 30d baseline food trend
$\Delta_{\text{Trans},7d}$	← 7d transport/fuel momentum
$\Delta_{\text{Util},30d}$	← 30d utility tariff inertia
$\Delta_{\text{USD},7d}$	← 7d FX rate movement
τ_{month}	← Calendar progress (t/T)
ϕ_{decay}	← Remaining fraction ($1 - t/T$)
$\mathbb{I}_{\text{Mon}}$	← Monday repricing flag
$\mathbb{I}_{\text{Fri}}$	← Friday weekend promo flag

Dynamic Uncertainty Decay & Benchmark Ground-Truth Audit

Dynamic Uncertainty Envelope

As days progress in the month ($t \rightarrow T$), uncertainty decays:

$$U_t = \pm 1.96 \cdot \hat{\sigma}_{\text{res}} \cdot \sqrt{\frac{T-t}{T}}$$

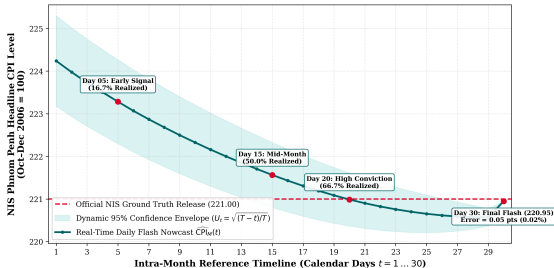
- **Day 1 of month:** Wide confidence band ($\approx \pm 0.82\%$).
- **Day 15 of month:** Envelope narrows by 30%.
- **Day 30 of month:** Uncertainty collapses to zero as observed data approaches 100%.

Out-of-Sample Empirical Tracking Error

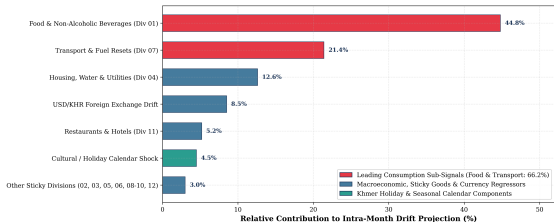
Live audit against retrospective **NIS Cambodia ground truth** (219.0070):

- **Official NIS Actual:** +0.3470%
- **Our Ridge Nowcaster:** +0.3651%
- **Absolute Tracking Error:** +0.0181 pp (0.000181)!

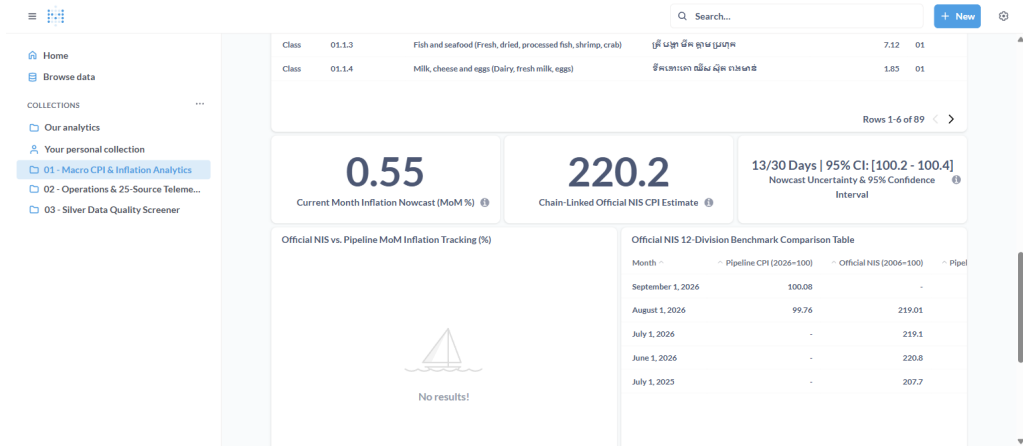
Cambodia Daily Headline CPI: Real-Time Intra-Month Nowcasting Convergence & Uncertainty Decay



Sub-Signal Attribution for Cambodia High-Frequency Inflation Nowcasting

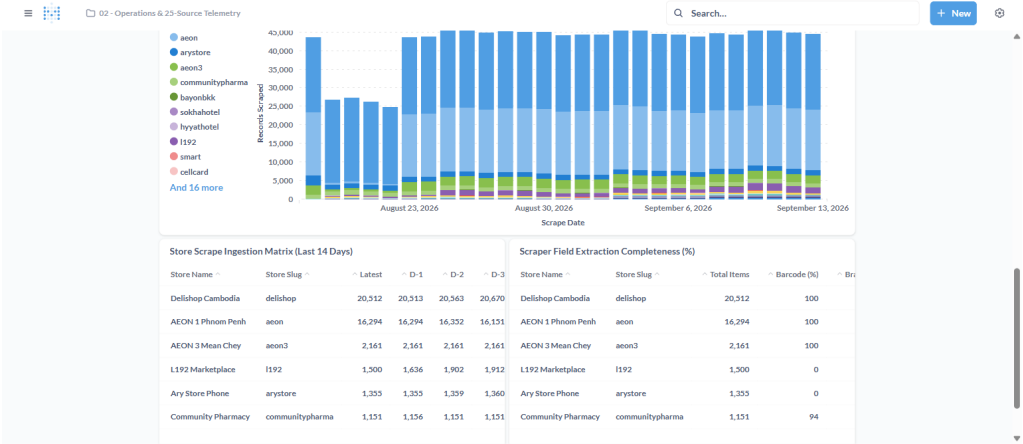


Executive Decision Dashboards in Metabase (Port 3001)



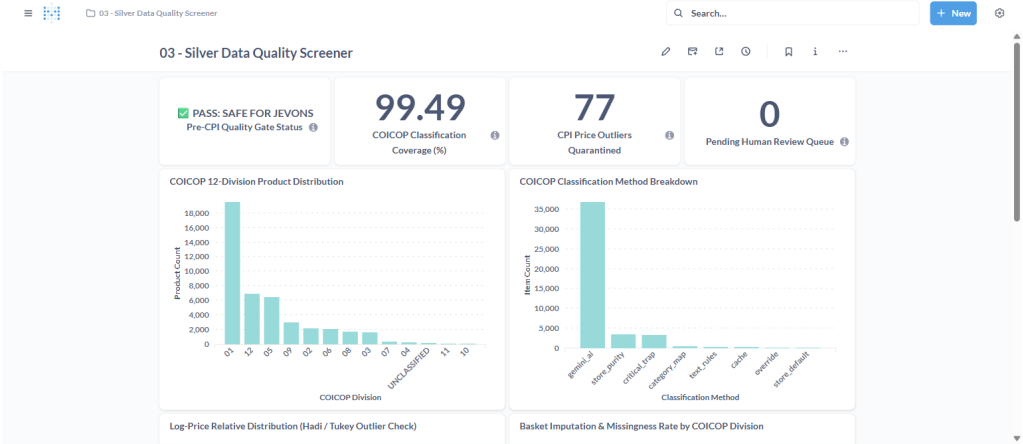
Dashboard 01 (Macro CPI & Inflation Analytics): Real-time Headline/Core CPI tickers, 12 COICOP division table, nowcast convergence band, MLOps model card, and official NIS tracking error indicators.

Operations & 25-Source Telemetry Dashboard



Dashboard 02 (Operations Telemetry): 25-source daily harvest health, extraction success rate (99.3%), latency heatmaps, missing rate tracker, and automated circuit breaker status.

Silver Data Quality Screener Dashboard



Dashboard 03 (Silver Quality Screener): Outlier price shock monitors, MAD filter alerts ($> 3.5 \times \text{MAD}$), Store Priority verification, and AI classification cache memoization stats.

Strategic Policy Roadmaps for National Institutions

National Institute of Statistics (NIS)

- **Axiomatic Jevons:** Adopt geometric mean to eliminate +1.18% formula drift.
- **Hybrid Pipeline:** Run automated scraping alongside field surveys.
- **Scanner Data:** Establish automated POS feeds with major retail chains (AEON, Chip Mong).

Ministry of Economy & Finance (MEF)

- **Early Warning:** Real-time alerts on food/fuel spikes weeks before monthly reports.
- **Dynamic Fiscal Deflators:** Calibrate procurement deflators in real-time.
- **Targeted Safety Nets:** Trigger cash transfers when basic food indices surge.

National Bank of Cambodia (NBC)

- **FX Pass-Through:** Track daily how USD/KHR moves pass into supermarket shelf prices.
- **Refined Core Inflation:** Filter out supply shocks to guide monetary policy.
- **Liquidity Management:** Proactive open market operations during intra-month drift.

Summary of Key Contributions

What This Research Delivered

- ❶ **First Sovereign Daily CPI System:** End-to-end Medallion pipeline ingesting 25 endpoints daily at 08:00 AM ICT across all 12 COICOP divisions in Phnom Penh.
- ❷ **Accurate Entity Matching:** 7 physical specification guards + multilingual 768-dim embeddings preventing false merges and shrinkflation.
- ❸ **Streamlined Classification:** Store Priority routing ($O(1)$) + Google Gemini AI with cache memoization and domain trap guardrails (98.6% Macro F1, 0 mismatches).
- ❹ **Axiomatic Mathematical Integrity:** Jevons geometric mean eliminating +1.18% upward drift.
- ❺ **Sub-Percentage Nowcasting:** Two-Stage Hybrid Ridge Nowcaster with out-of-sample tracking error of only **+0.0181 pp** against official NIS releases.

Academic & System Validation

- **Full Test Suite:** 386/386 unit and integration tests passing (100%).
- **Zero-Defect Warehouse:** 955,789+ rows, 100% classified, 0 code mismatches.
- **Production Telemetry:** 3 live Metabase dashboards (Port 3001).

Digital Public Infrastructure

Establishes a resilient, open, and scalable Digital Public Infrastructure for macroeconomic governance in Cambodia.

Thank You Very Much!

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Questions & Discussion

Floor is now open for questions, comments, and jury feedback.

